Hamiltonian Monte Carlo for Hierarchical Models

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Hierarchical Bayesian Models

Consider a one-way normal hierarchical model

$$\pi(\theta, \phi \mid D) \propto \prod_{i=1}^{I} \pi(D_i \mid \theta_i)\pi(\theta_i \mid \phi)\pi(\phi).$$

with

$$y_i \sim N(\theta_i, \sigma_i^2),$$

$$\theta_i \sim N(\mu, \tau^2), \qquad i = 1, \ldots, I.$$

Define

$$D = \{y_i, \sigma_i\}, \qquad \phi = (\mu, \tau), \qquad \theta = (\theta_i).$$

The Funnel Geometry

Key observation

Small changes in the hyperparameters can induce large changes in the posterior density.

When data are weakly informative, hierarchical posteriors often exhibit a funnel geometry.

$$\pi(\theta_1, \dots, \theta_n, v) \propto \prod_{i=1}^{n} N(\theta_i \mid 0, e^{-v/2}) N(v \mid 0, 3^2).$$

- The curvature varies strongly across the support.
- Local geometry changes dramatically.
- Sampling becomes difficult.

Figure 2: Funnel Geometry

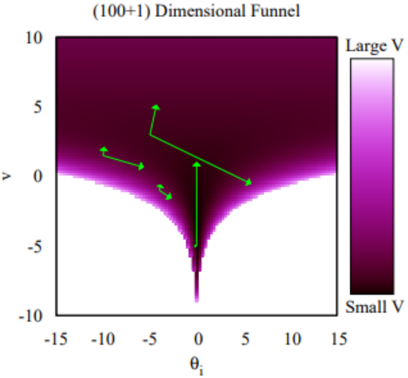


FIG. 2. Typical of hierarchical models, the curvature of the funnel distribution varies strongly with the parameters, taxing most algorithms and limiting their ultimate performance. Here the curvature is represented visually by the eigenvectors of $\sqrt{|\partial^2 \log p(\theta_1, \ldots, \theta_n, v) / \partial\theta_i \partial\theta_j|}$ scaled by their respective eigenvalues, which encode the direction and magnitudes of

The eigenvectors of the Hessian encode

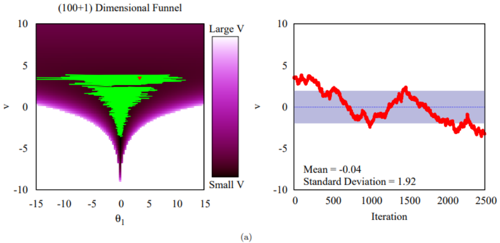
- local directions of curvature,
- local magnitudes of curvature,
- deviation from isotropy.

Main message

The difficulty is not simply global correlation; the geometry changes across the parameter space.

Why Gibbs and Metropolis Struggle

- Hierarchical posteriors induce strong dependencies.
- Naive implementations become inefficient.
- Exploration can degenerate into random-walk behavior.



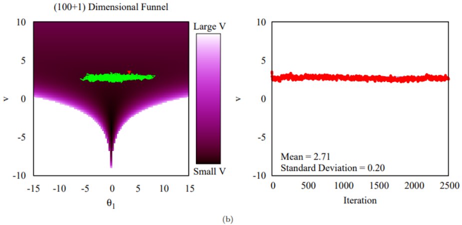


FIG. 3. One of the biggest challenges with modern models is not global correlations but rather local correlations that are resistant to the corrections based on the global covariance. The funnel distribution, here with $N = 100$, features strongly varying local correlations but enough symmetry that these correlations cancel globally, so no single correction can compensate for the ineffective exploration of (a) the Gibbs sampler and (b) Random Walk Metropolis. After 2500 iterations neither chain has explored the marginal distribution of v, $\pi(v) = \mathcal{N}(v|0, 3^2)$. Note that the Gibbs sampler utilizes a Metropolis-within-Gibbs scheme as the conditional $\pi(v|\vec{\theta})$ does not have a closed-form.

Figure 3

Both algorithms eventually exhibit random-walk behavior and fail to efficiently explore the funnel geometry.

Efficient Implementations

The paper discusses two strategies:

1. Parameter expansion.
2. Reparameterization.

Goal:

reduce posterior dependence

and improve computational efficiency.

Parameter Expansion

Introduce auxiliary variables:

$$\theta_i = \mu + \xi \eta_i,$$

$$\eta_i \sim N(0, \sigma_\eta^2),$$

$$\tau = |\xi| \sigma_\eta.$$

Advantages:

- Conditional independence may improve Gibbs.
- Hierarchical structure becomes more explicit.

Disadvantage:

- Additional multiplicative variables may worsen Metropolis-Hastings.

Non-Centered Parameterization

Instead of adding new parameters, reparameterize existing ones.

$$\theta_i = \mu + \tau \vartheta_i,$$

$$\vartheta_i \sim N(0, 1).$$

Equivalent model:

$$y_i \sim N(\mu + \tau \vartheta_i, \sigma_i^2).$$

Interpretation

Dependence is shifted from latent parameters toward the data model.

Centered vs Non-Centered

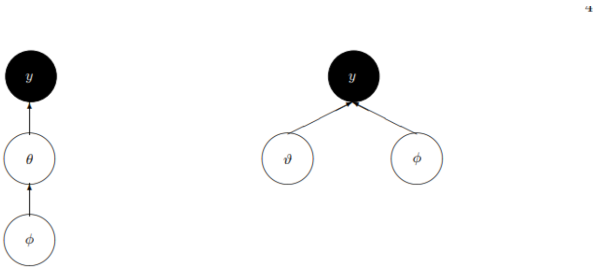


FIG. 4. In one-level hierarchical models with global parameters, ϕ, local parameters, θ, and measured data y, correlations between parameters can be mediated by different parameterizations of the model. Non-centered parameterizations exchange a direct dependence between ϕ and θ for a dependence between ϕ and y; the reparameterized ϑ and ϕ become independent conditioned on the data. When the data are weak these non-centered parameterizations yield simpler posterior geometries.

- Centered parameterizations perform better when data are highly informative.($\sigma << \tau$)
- Non-centered parameterizations perform better when data are weakly informative.($\sigma >> \tau$)

Hamiltonian Monte Carlo

- Exploit differential geometry to eliminate the random-walk behavior of MH and Gibbs.
- Introduce auxiliary momentum variables p to the target distribution q. Construct the augmented distribution:
$$\pi(p, q) = \pi(p \mid q)\pi(q).$$
- The Hamiltonian is
$$\begin{aligned} H(p, q) &= -\log \pi(p, q) \\ &= -\log \pi(p \mid q) - \log \pi(q). \end{aligned}$$

Potential and Kinetic Energy

Define

$$
\begin{aligned}
T(p \mid q) &= -\log \pi(p \mid q) \quad \text{(Kinetic Energy)} \\
V(q) &= -\log \pi(q) \qquad\quad\ \ \text{(Potential Energy)}
\end{aligned}
$$

Then

$$H(p, q) = T(p \mid q) + V(q).$$

Hamilton's Equations

Sample momentum

$$p \sim \pi(p \mid q).$$

Then evolve the joint system according to Hamilton's equations:

$$\frac{dq}{dt} = +\frac{\partial H}{\partial p} = +\frac{\partial T}{\partial p},$$

$$\frac{dp}{dt} = -\frac{\partial H}{\partial q} = -\frac{\partial T}{\partial q} - \frac{\partial V}{\partial q}.$$

The gradients guide transitions through regions of high posterior probability.

Advantages of HMC

Efficient HMC transitions persist even in correlated target distributions.

However, performance depends critically on the choice of

$$\pi(p \mid q).$$

Euclidean HMC

Choose a Gaussian momentum distribution independent of q:

$$\pi(p \mid q) = N(p \mid 0, \Sigma).$$

The resulting kinetic energy is quadratic:

$$T(p, q) = \frac{1}{2} p^\top \Sigma^{-1} p.$$

Interpretation

The covariance matrix Σ induces a global metric on the parameter space.

Weaknesses of Euclidean HMC

Two main limitations:

1. Characteristic length scale.
2. Limited density variation.

These become severe in hierarchical models.

Limitation I: Characteristic Length Scale

Hamilton's equations are usually too complex to solve analytically, so the trajectory must be integrated numerically.

The numerical integrator introduces a characteristic length scale: the step size ϵ.
For a stable numerical solution, the step size must match the local curvature:

$$\epsilon \sqrt{\lambda_i} < 2,$$

for each eigenvalue λ_i of the matrix

$$M_{ij} = (\Sigma^{-1})_{ik} \frac{\partial^2 V}{\partial q_k \partial q_j}.$$

Consequence

In hierarchical models, curvature varies strongly across the posterior. A step size that is stable in one region may be inefficient or unstable in another.

Limitation II: Limited Density Variation

Hamiltonian dynamics preserve

$$H(p, q) = T(p \mid q) + V(q).$$

In Euclidean HMC,

$$\Delta V = \Delta T \sim \frac{d}{2}.$$

- Hierarchical models induce huge density variations.
- Energy variation is limited.
- Random-walk behavior reappears.

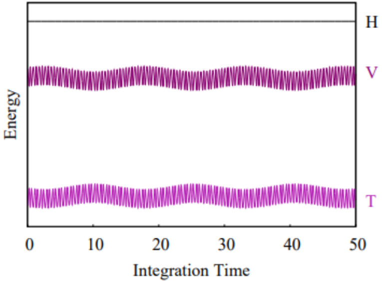


FIG. 6. Because the Hamiltonian, H, is conserved during each trajectory, the variation in the potential energy, V, is limited to the variation in the kinetic energy, T, which itself is limited to only $d/2$.

Non-Centered Parameterization Helps

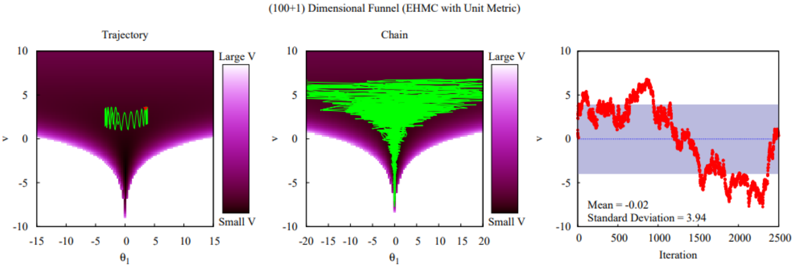
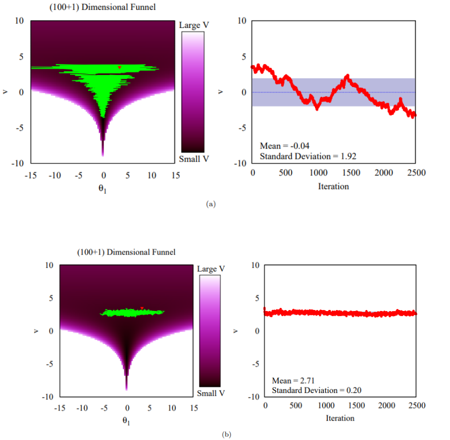


FIG. 7. Limited to moderate potential energy variations, the trajectories of Euclidean HMC, here with a unit metric $\Sigma = \mathbb{I}$, reduce to random walk behavior in hierarchical models. The resulting Markov chain explores more efficiently than Gibbs and Random Walk Metropolis (Figure 3), but not efficiently enough to make these models particularly practical.

As a consequence:

- Euclidean HMC becomes more efficient.
- Funnel geometries become less severe.

Non-Centered Parameterization Helps



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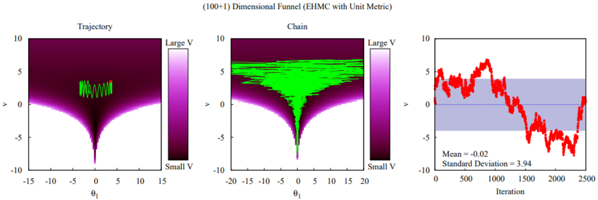

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Centered versus Non-Centered

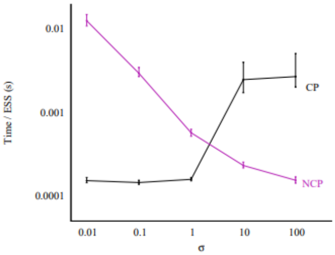


FIG. 8. Depending on the common variance, σ^2, from which the data were generated, the performance of a 10-dimensional one-way normal model (2) varies drastically between centered (CP) and non-centered (NCP) parameterizations of the latent parameters, θ_i. As the variance increases and the data become effectively more sparse, the non-centered parameterization yields the most efficient inference and the disparity in performance increases with the dimensionality of the model. The bands denote the quartiles over an ensemble of 50 runs, with each run using Stan [20] configured with a diagonal metric and the No-U-Turn sampler. Both the metric and the step size were adapted during warmup, and care was taken to ensure consistent estimates (Figure 9).

Performance depends on

$$\sigma^2.$$

- Small variance favors centered parameterizations.
- Large variance favors non-centered parameterizations.

Centered versus Non-Centered

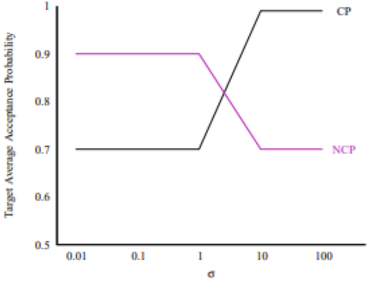


FIG. 9. As noted in Section III A 1, care must be taken when using adaptive implementations of Euclidean Hamiltonian Monte Carlo with hierarchical models. For the results in Figure 8, the optimal average acceptance probability was relaxed until the estimates of τ stabilized, as measured by the potential scale reduction factor, and divergent transitions did not appear.

Riemannian HMC

Euclidean HMC can be generalized by allowing the covariance matrix to vary with position:

$$\Sigma = \Sigma(q).$$

Then the momentum distribution becomes

$$\pi(p \mid q) = N(p \mid 0, \Sigma(q)).$$

The kinetic energy is

$$T(p, q) = \frac{1}{2} p^\top \Sigma^{-1}(q) p - \frac{1}{2} \log |\Sigma(q)|.$$

Interpretation

The metric adapts locally to the geometry of the target distribution.

Riemannian HMC and Metric Choice

Euclidean HMC can be generalized by allowing the covariance matrix to vary with position:

$$\Sigma = \Sigma(q).$$

Then

$$\pi(p \mid q) = N(p \mid 0, \Sigma(q)).$$

and

$$T(p, q) = \frac{1}{2} p^\top \Sigma^{-1}(q) p - \frac{1}{2} \log |\Sigma(q)|.$$

Advantages:

- Local adaptation to posterior geometry.
- Better exploration of funnel distributions.
- Larger density variations become accessible.

A natural choice is the Hessian of the potential energy.

Riemannian HMC and Metric Choice

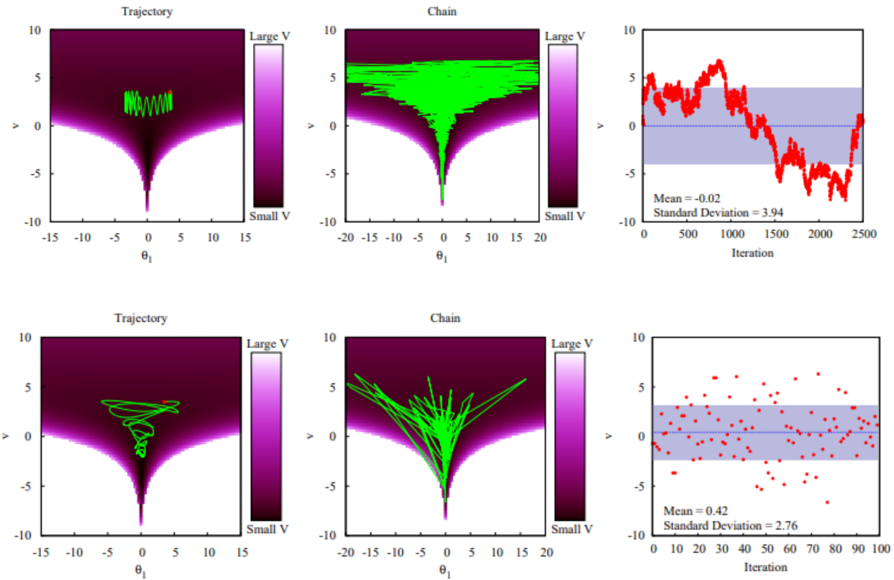
However:

- Hessians are not always positive definite.
- Hessians may be numerically unstable.

Solution

The paper proposes the **SoftAbs metric**, a smooth positive-definite approximation based on the Hessian eigenvalues.

Riemannian HMC and Metric Choice



Example: One-Way Normal Model

The paper considers

$$I = 800, \qquad \sigma_i = 10, \qquad \mu = 8, \qquad \tau = 3.$$

with priors

$$\mu \sim N(0, 5^2), \qquad \tau \sim \text{Half-Cauchy}(0, 2.5).$$

Centered

$$y_i \sim N(\theta_i, \sigma_i^2),$$

$$\theta_i \sim N(\mu, \tau^2).$$

Non-centered

$$y_i \sim N(\mu + \tau\vartheta_i, \sigma_i^2),$$

$$\vartheta_i \sim N(0, 1).$$

Results

Algorithm	**Parameterization**	**Step Size**	**Average Acceptance Probability**	**Time (s)**	**Time/ESS (s)**
Metropolis	Centered	$5.00 \cdot 10^{-3}$	0.822	$4.51 \cdot 10^{4}$	1220
Gibbs	Centered	1.50	0.446	$9.54 \cdot 10^{4}$	297
EHMC	Centered	$1.91 \cdot 10^{-2}$	0.987	$1.00 \cdot 10^{4}$	**16.2**
Metropolis	Non-Centered	0.0500	0.461	398	1.44
Gibbs	Non-Centered	2.00	0.496	817	1.95
EHMC	Non-Centered	0.164	0.763	154	$\mathbf{2.94 \cdot 10^{-2}}$

TABLE I. Euclidean Hamiltonian Monte Carlo significantly outperforms both Random Walk Metropolis and the Metropolis-within-Gibbs sampler for both parameterizations of the one-way normal model. The difference is particularly striking for the more efficient non-centered parameterization that would be used in practice.

Conclusions

1. Hierarchical models induce difficult geometries.
2. Curvature varies strongly across the posterior.
3. Euclidean HMC improves exploration but remains limited.
4. Non-centered parameterizations can substantially improve performance.
5. Riemannian HMC adapts to local geometry.
6. SoftAbs provides a practical metric construction.